

Laboratory Work 1 — ERD Diagram

System: International Airport Database

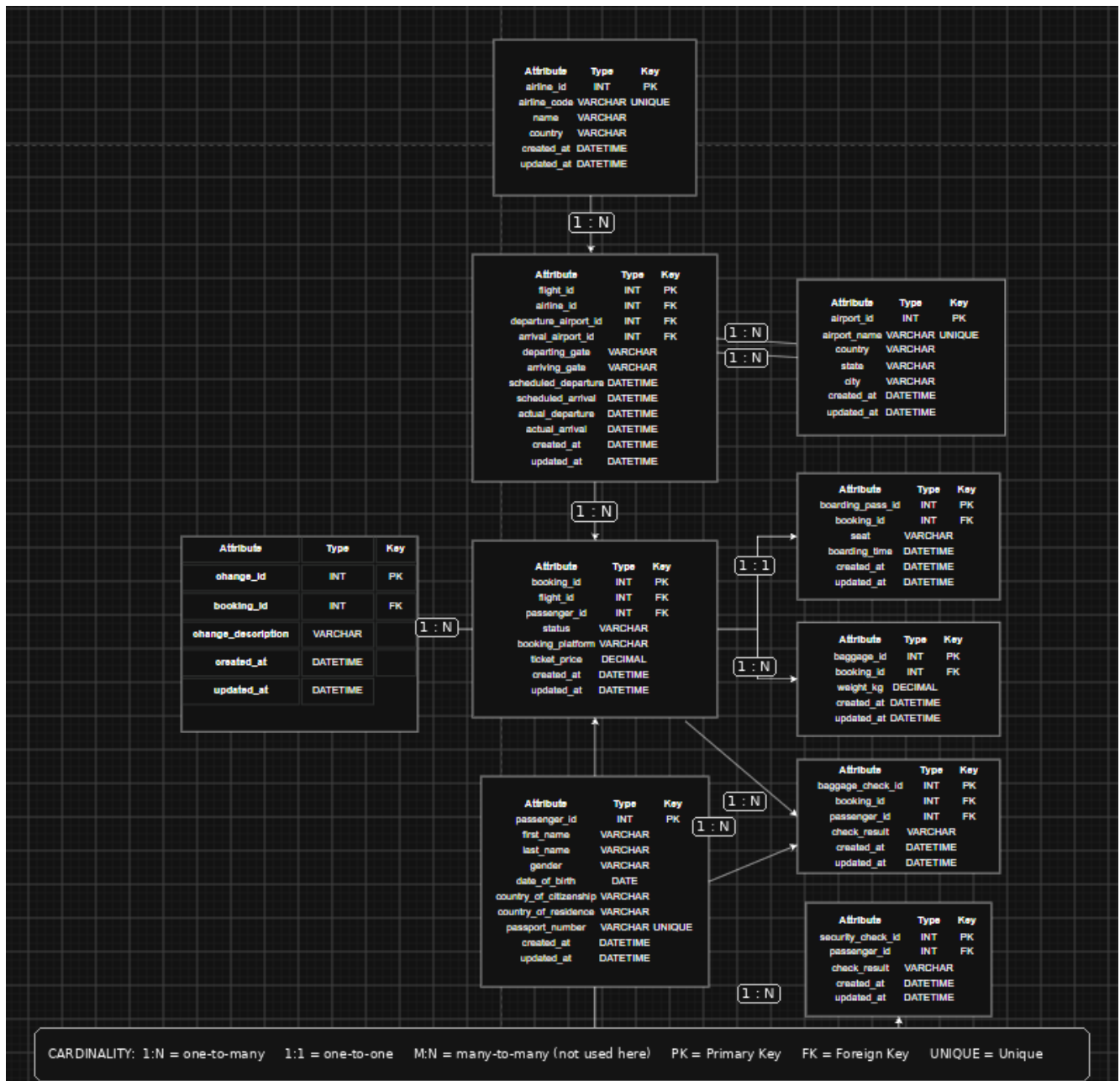


Figure 1. Entity-Relationship Diagram of the International Airport system. Relationship labels show the main cardinalities.

1. Description of the Solution

Airport: Stores airport information: `airport_id`, `airport_name`, `country`, `state`, `city`, `created_at` and `updated_at`.

Airline: Stores airline information: `airline_id`, `airline_code`, `name`, `country`, `created_at` and `updated_at`.

Flight: Stores scheduled and actual flight information, including `airline_id`, departure and arrival airport IDs, gates, scheduled times, actual times, `created_at` and `updated_at`.

Booking: Represents a passenger booking for a flight. It contains `booking_id`, `flight_id`, `passenger_id`, `status`, `booking_platform`, `ticket_price`, `created_at` and `updated_at`.

Passenger: Stores passenger profile information such as name, gender, date of birth, citizenship, residence, passport number, `created_at` and `updated_at`.

Booking Change: Stores changes made to a booking using `change_id`, `booking_id`, `change_description`, `created_at` and `updated_at`.

Boarding Pass: Stores boarding pass information including `boarding_pass_id`, `booking_id`, `seat`, `boarding_time`, `created_at` and `updated_at`.

Baggage: Stores baggage registered for a booking: `baggage_id`, `booking_id`, `weight_kg`, `created_at` and `updated_at`.

Baggage Check: Stores baggage-check information including `baggage_check_id`, `booking_id`, `passenger_id`, `baggage_id`, `check_result`, `created_at` and `updated_at`.

Security Check: Stores passenger security-check information including `security_check_id`, `passenger_id`, `check_result`, `created_at` and `updated_at`.

2. Relationships and Cardinality

Relationship	Cardinality	Explanation
Airline → Flight	1:N	One airline can be associated with many flights; each flight references one airline.
Airport → Flight (departure)	1:N	One airport can be the departure airport for many flights; each flight references one departure airport.
Airport → Flight (arrival)	1:N	One airport can be the arrival airport for many flights; each flight references one arrival airport.
Flight → Booking	1:N	One flight can have many bookings; each booking references one flight.
Passenger → Booking	1:N	One passenger can have many bookings; each booking references one passenger.
Booking → Booking Change	1:N	One booking can have multiple recorded changes; each change references one booking.
Booking → Boarding Pass	1:1	A booking is associated with a boarding pass in the simplified model.
Booking → Baggage	1:N	One booking can have multiple baggage records; each baggage record references one booking.
Booking/Passenger → Baggage Check	1:N	Baggage-check records are associated with a booking and passenger; the FK fields maintain those links.
Passenger → Security Check	1:N	A passenger can have security-check records; each security check references one passenger.

3. Notation and Participation Constraints

PK — Primary Key; FK — Foreign Key; UNIQUE — unique attribute. 1:N means one-to-many and 1:1 means one-to-one. The diagram uses the FK structure to show which child record references a parent record. The source task does not explicitly specify NULL/NOT NULL constraints for all foreign keys, so exact optional/mandatory participation is not fully defined by the provided specification. For that reason, the diagram focuses on the supported cardinalities rather than inventing NOT NULL requirements.

There is no direct M:N relationship in the presented schema; the provided entities are connected through foreign-key relationships.

4. Conclusion

The ERD represents the airport system using separate entities for airports, airlines, flights, bookings, passengers, booking changes, boarding passes, baggage, baggage checks and security checks. Primary and foreign keys maintain the relationships between the entities, while the cardinality labels describe the main one-to-many and one-to-one relationships.